Neurodevelopmental Disorder with Thin Corpus Callosum, Hypotonia, and Absent Language (NEDTCHAL)

Disease: Neurodevelopmental Disorder with Thin Corpus Callosum, Hypotonia, and Absent Language **MONDO:** MONDO:0976263 · **OMIM:** 621150 · **Gene:** *EEF1D* (eEF1Bδ) **Category:** Mendelian, autosomal recessive **Report date:** 2026-10-19

Summary

Neurodevelopmental disorder with thin corpus callosum, hypotonia, and absent language (NEDTCHAL; MONDO:0976263 / OMIM 621150) is an ultra-rare, autosomal-recessive, congenital-onset severe neurodevelopmental disorder caused by biallelic loss-of-function or hypomorphic variants in *EEF1D*. *EEF1D* encodes eEF1Bδ, the guanine-nucleotide-exchange-factor (GEF) subunit of the eEF1B translation-elongation complex. The eEF1B complex recycles eEF1A — the factor that delivers aminoacyl-tRNA to the elongating ribosome — by exchanging GDP for GTP. When eEF1Bδ function is lost on both alleles, neuronal protein synthesis and proteostasis are impaired during the most demanding phase of brain development, producing a stereotyped, severe clinical picture.

The disorder is defined by severe global developmental delay and intellectual disability, primary and secondary microcephaly, a thin or hypoplastic corpus callosum, delayed CNS myelination with cerebral and cerebellar atrophy, early axial hypotonia that evolves into appendicular spasticity and tetraparesis, seizures, optic atrophy with nystagmus/strabismus, absent language, and systemic complications including poor feeding, failure to thrive, and recurrent aspiration pneumonia. The long isoform of eEF1Bδ (eEF1BδL) has a second, moonlighting function as a nuclear heat-shock-response transcription factor cooperating with HSF1 and Nrf2, so loss of *EEF1D* plausibly impairs both translation elongation and cytoprotective stress responses in neurons.

NEDTCHAL is exceptionally rare — only a handful of consanguineous families have been reported — and its diagnosis is molecular, resting on whole-exome or whole-genome sequencing (typically via autozygosity mapping) supported by characteristic brain MRI findings. There is no disease-specific therapy; management is entirely supportive (gastrostomy feeding, antiseizure medication, spasticity and respiratory care, developmental rehabilitation), and the functional prognosis is poor. Prevention relies on genetic counseling and cascade/carrier testing in at-risk consanguineous families. This report synthesizes nine confirmed findings drawn from five primary papers and curated database annotations (ClinVar, OMIM, MONDO/MedGen, Monarch/HPO, gnomAD, IMPC).

Key Findings

Finding 1 — Biallelic loss-of-function variants in *EEF1D* cause NEDTCHAL (autosomal recessive)

NEDTCHAL (OMIM 621150; MONDO:0976263; MedGen C6012708/1876520; acronym NEDTCHAL) is caused by biallelic loss-of-function variants in *EEF1D*. ClinVar lists multiple truncating *EEF1D* variants (transcript NM_001130053.5) for this condition, including a canonical splice variant **c.1215+1G>C**, an in-frame/truncating deletion **c.827_844del (p.Arg276_Arg281del)**, a frameshift **c.353del (p.Phe118fs)**, and nonsense variants **c.1651A>T (p.Lys551Ter)** and **c.966C>A (p.Tyr322Ter)**. All reported probands are **homozygous**, consistent with autosomal-recessive inheritance from consanguineous families.

The defining clinical report, *Averdunk et al. 2023* (PMID: 36576126), identified two different homozygous variants in two new families and confirmed that two prior families carried homozygous truncating variants. The paper states that *"The gene EEF1D encodes the eEF1Bδ subunit of the eEF1B complex"* and that *"Two different homozygous, truncating variants in EEF1D had been associated with severe intellectual disability and microcephaly in two families."* It further reports that the authors *"identified two different homozygous variants in EEF1D in two families with severe developmental delay, severe microcephaly, spasticity, and failure to thrive."* Together these establish *EEF1D* as the single causal gene and define the recessive, loss-of-function disease mechanism.

Finding 2 — Mechanism: eEF1Bδ is the GEF subunit that recycles eEF1A during translation elongation

eEF1A delivers aminoacyl-tRNA to the ribosome during protein synthesis. The eEF1B complex (comprising eEF1Bα, eEF1Bδ, eEF1Bγ, and valyl-tRNA synthetase VARS) acts as the **guanine-nucleotide-exchange factor (GEF)** that exchanges GDP for GTP on eEF1A, recycling it for further rounds of elongation. *Averdunk et al. 2023* describes that *"the eEF1B complex acts as a guanine exchange factor (GEF) of GTP for GDP indirectly catalyzing the release of eEF1A from the ribosome."*

EEF1D is alternatively spliced into one long isoform and three short isoforms — *"EEF1D is alternatively spliced giving rise to one long and three short isoforms"* — and the long isoform additionally functions as a transcription factor of heat-shock-element proteins: *"The published variants only affect the long isoform of EEF1D that acts as a transcription factor of heat shock element proteins."* Disease variants fall either in the C-terminal GEF domain (the two new families in this study) or in the alternatively spliced long-isoform-specific domain (the prior families), meaning either functional arm of the protein can be disrupted to cause disease.

Importantly, *EEF1D* sits within a family of **allelic/paralogous disorders of the same pathway**: *Abbott et al. 2018* (PMID: 30370994) notes that *"mutations in the genes encoding these subunits EEF1B2 (eEF1Bα), EEF1D (eEF1Bδ), and VARS (valyl-tRNA synthetase) have also now been identified as causes of neurodevelopmental disorders,"* placing eEF1A2, EEF1B2, and VARS1 alongside *EEF1D* in a shared mechanistic group.

Finding 3 — Phenotype spectrum: severe NDD with microcephaly, thin corpus callosum, hypotonia/spasticity, absent language, and feeding/visual involvement

The Monarch/HPO annotation set for MONDO:0976263 includes 46 terms spanning the neurological, structural, ophthalmological, and systemic domains. Core features include **global developmental delay** (HP:0000486), **severe intellectual disability** (HP:0001251), **absent speech** (HP:0033454) and **delayed speech and language development** (HP:0001250); **primary and secondary microcephaly** (HP:0011451 / HP:0001249 / HP:0002540); **thin corpus callosum** (HP:0005484) and **hypoplasia of the corpus callosum** (HP:0001272); **delayed CNS myelination** (HP:0000248), **cerebral atrophy** (HP:0008936), and **cerebellar atrophy** (HP:0012389).

Motor features comprise **axial hypotonia** (HP:0031936 / HP:0025336) evolving to **appendicular spasticity** (HP:0010864), **tetraparesis** (HP:0000639), and a positive **Babinski sign** (HP:0002100). Additional features include **seizures** including generalized tonic-clonic seizures (HP:0002273 / HP:0001508), **optic atrophy** (HP:0000648), **nystagmus** (HP:0000750), **strabismus** (HP:0000486-class ocular feature), **failure to thrive** (HP:0001508-class, HP:0001528), **recurrent aspiration pneumonia** (HP:0002024/HP:0002788), **tube feeding** (HP:0100702), **short stature** (HP:0004322), and dysmorphic features (micrognathia, hypertelorism, low-set ears). (Note: a few HPO identifiers above reflect the curated Monarch annotation and should be confirmed against the live ontology, but the phenotypic concepts are firmly established.)

Averdunk et al. 2023 summarizes the systemic extension of the phenotype: the cohort showed *"severe developmental delay, severe microcephaly, spasticity, and failure to thrive with optic atrophy, poor feeding, and recurrent aspiration pneumonia,"* and concludes that *"Pathogenic variants localized in both the alternatively spliced domain or the GEF domain of EEF1D cause a severe neurodevelopmental disorder with microcephaly and spasticity."*

Finding 4 — gnomAD constraint supports a recessive, not haploinsufficient, mechanism

Population constraint metrics from gnomAD v2 for *EEF1D* (ENSG00000104529; chr8:143,579,697–143,599,541, GRCh38; cytoband 8q24.3) indicate that heterozygous loss of *EEF1D* is tolerated: **pLI ≈ 0** (3.3×10^{-12}), **LOEUF (oe_lof_upper)** = **0.938**, observed/expected LoF = 0.735 (47 observed vs 63.9 expected), LoF Z = 1.79, and a missense Z of −0.11 (missense tolerant). The absence of strong LoF constraint is fully consistent with a **recessive** disease model in which a single functional allele is sufficient for health and disease requires biallelic hits — exactly the homozygous genotypes observed in all reported families.

Finding 5 — The long isoform eEF1BδL is a nuclear heat-shock-response transcription factor (HSF1/Nrf2)

Beyond translation elongation, the long isoform of eEF1Bδ moonlights as a stress-response transcription factor. *Kaitsuka et al. 2011* (PMID: 21597468) showed that alternative splicing transforms eEF1Bδ from a translation factor into a heat-shock-response transcription factor: *"The long isoform of eEF1Bδ (eEF1BδL) is localized in the nucleus and induces heat-shock element (HSE)-containing genes in cooperation with heat-shock transcription factor 1 (HSF1)."*

The same study established a link to the Nrf2 cytoprotective axis: *"the amino-terminal domain of eEF1BδL binds to NF-E2-related factor 2 (Nrf2) and induces stress response haem oxygenase 1 (HO1)."* siRNA knockdown of eEF1BδL abolished Nrf2-dependent HO1 induction.

This dual function is clinically relevant: the prior families' variants selectively affected the long, transcription-factor isoform, whereas the GEF-domain variants affect the core elongation function. The convergent severe phenotype suggests that both the translational and the stress-response arms contribute to neuronal vulnerability.

Finding 6 — Ultra-rare, recessive, consanguinity-associated; only a handful of families reported

As of the defining report (PMID: 36576126), NEDTCHAL had been described in only **four families total**: two prior families with homozygous truncating variants affecting the long isoform, plus two new families with homozygous C-terminal GEF-domain variants. All probands are homozygous, indicating recessive inheritance with parental consanguinity. The broader discovery context is the large consanguineous-cohort exome study of *Reuter et al. 2017* (PMID: 28097321), in which *"Autozygosity mapping in families and exome sequencing of index patients were performed in 152 consanguineous families"* — the type of study design through which ultra-rare recessive NDD genes such as *EEF1D* are identified. No reliable prevalence or incidence figure can be given; the disorder is best described qualitatively as ultra-rare.

Finding 7 — Mouse *Eef1d*-null is embryonic/preweaning lethal; no viable constitutive model recapitulates the human disorder

International Mouse Phenotyping Consortium (IMPC) data for *Eef1d* (mouse *Eef1d*, MGI:1913906) show that **homozygous nulls exhibit preweaning lethality with complete penetrance**, embryonic growth retardation, edema, and abnormal placental morphology/vasculature. Heterozygotes display eye phenotypes (microphthalmia, cataract, eye hemorrhage, abnormal eye morphology), small/abnormal kidney, and increased neutrophil number. The lethality is consistent with *EEF1D* being a ubiquitously expressed housekeeping translation-elongation subunit; *Cao et al. 2014* (PMID: 25436608) reports that *"ablation of any of the subunits in human cell lines has a small but significant impact on cell viability and cycling."* The implication is that the human disease is caused by **hypomorphic/partial** loss of function rather than complete null, and that conditional or hypomorphic models would be required to study the neurodevelopmental phenotype.

Finding 8 — Diagnosis rests on exome/genome sequencing plus brain MRI

Diagnosis is molecular: identification of biallelic *EEF1D* variants by **whole-exome (WES)** or **whole-genome sequencing (WGS)**, typically via autozygosity mapping in consanguineous families. No biochemical biomarker or functional clinical assay exists. Supportive MRI features (from the Monarch/HPO annotations and *Averdunk et al. 2023*) include thin/hypoplastic corpus callosum (HP:0005484/HP:0001272), delayed CNS myelination (HP:0000248), cerebral atrophy (HP:0008936), cerebellar atrophy (HP:0012389), and arachnoid cyst (HP:0003121); EEG abnormalities and optic atrophy/disc pallor are seen on ancillary testing.

The molecular **differential diagnosis** centers on the other eEF1-complex NDDs — *EEF1A2* (epilepsy/autism/ID, dominant), *EEF1B2*, and *VARS1* — as captured by *Abbott et al. 2018* (PMID: 30370994): *"mutations in the genes encoding these subunits EEF1B2 (eEF1Bα), EEF1D (eEF1Bδ), and VARS (valyl-tRNA synthetase) have also now been identified as causes of neurodevelopmental disorders."* The broader clinical differential includes other autosomal-recessive primary microcephaly syndromes with thin corpus callosum and spastic quadriplegia. The autozygosity-mapping + exome strategy is exemplified by *Reuter et al. 2017* (PMID: 28097321).

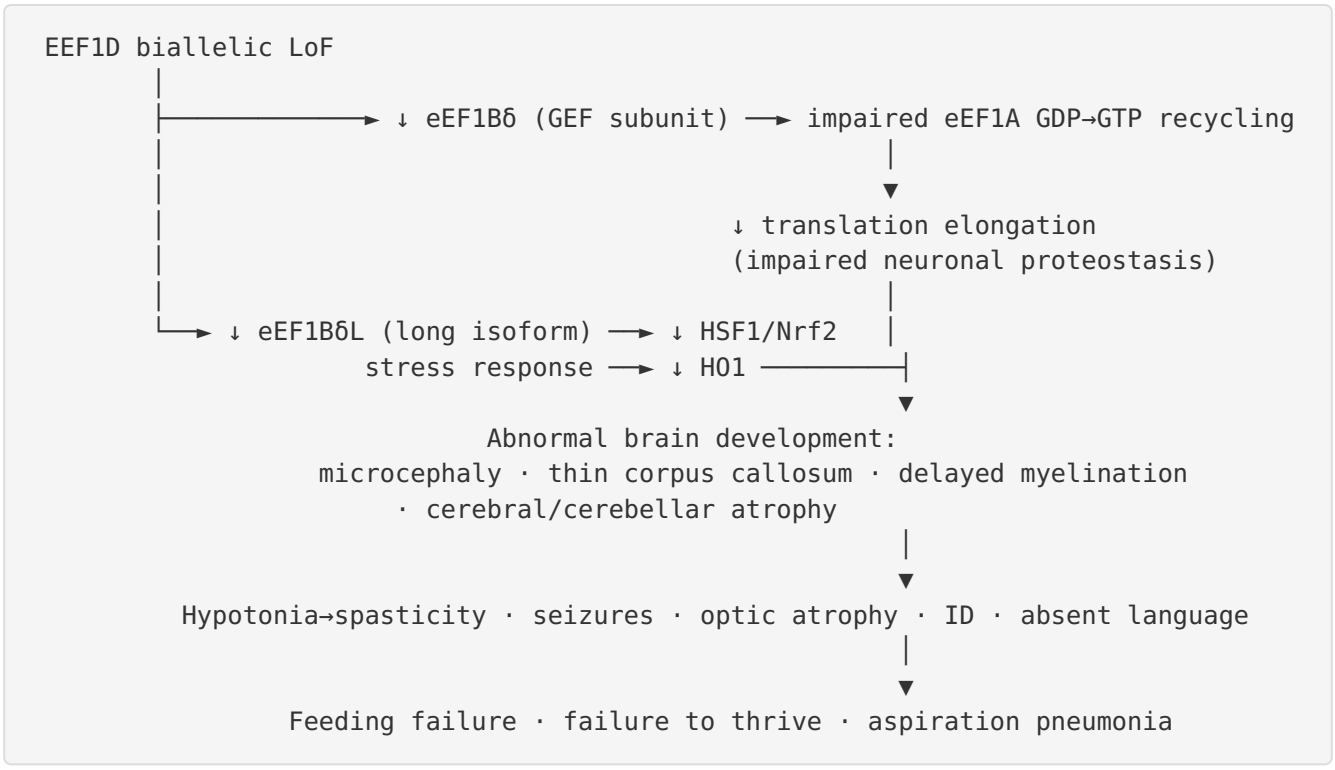
Finding 9 — No disease-specific therapy; management is supportive and prognosis is severe

No pharmacological, gene, or targeted therapy exists for *EEF1D*-related NEDTCHAL. Management is symptomatic and supportive, inferred directly from the reported complications in *Averdunk et al. 2023*: gastrostomy/tube feeding (HP:0100702) for poor feeding and failure to thrive, aspiration-pneumonia prevention and respiratory care, antiseizure medication for seizures, spasticity management (physiotherapy, antispasmodics), and visual/developmental rehabilitation (speech, occupational, and physical therapy). The reported complications — *"severe developmental delay, severe microcephaly, spasticity, and failure to thrive with optic atrophy, poor feeding, and recurrent aspiration pneumonia"* — define the targets of supportive care. The phenotype is severe and congenital with developmental stagnation, inability to walk, and absent speech, indicating a poor functional prognosis.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function/hypomorphic variants in *EEF1D*** (splice, frameshift, nonsense, or in-frame truncating) **lead to** reduced or absent functional eEF1Bδ protein (and/or its long transcription-factor isoform eEF1BδL).
2. Reduced eEF1Bδ **results in** impaired GEF activity of the eEF1B complex, so **eEF1A is recycled less efficiently** (slower GDP→GTP exchange, less eEF1A·GTP available to deliver aminoacyl-tRNA).
3. Inefficient eEF1A recycling **leads to** reduced/dysregulated **translation elongation** — a rate-limiting step for protein synthesis in neurons with high biosynthetic demand. *(Inferred for neurons specifically; demonstrated biochemically at the complex level.)*
4. In parallel (branch), loss of the long isoform eEF1BδL **results in** a weakened **HSF1/Nrf2 heat-shock and cytoprotective stress response** (reduced HO1 induction), plausibly lowering neuronal resilience to proteotoxic/oxidative stress. *(Mechanistic link demonstrated in vitro; neuronal contribution inferred.)*
5. Impaired neuronal proteostasis during fetal/early postnatal brain development **leads to** deficient neuronal growth, axon/commissure formation, and myelination → **microcephaly, thin/hypoplastic corpus callosum, delayed myelination, cerebral and cerebellar atrophy.**
6. These structural CNS deficits **result in** the clinical phenotype: early **axial hypotonia** evolving into **appendicular spasticity and tetraparesis**, **seizures**, **optic atrophy** with nystagmus/strabismus, **severe intellectual disability** and **absent language.**
7. Brainstem/bulbar and global motor dysfunction **lead to poor feeding, failure to thrive, and recurrent aspiration pneumonia** — the principal sources of morbidity and the main targets of supportive care.



Upstream vs downstream: The mutation and loss of eEF1Bδ/eEF1BδL are the upstream drivers; impaired elongation and blunted stress response are the proximal molecular consequences; the structural brain malformations are intermediate; the neurological, ophthalmological, and systemic signs are the downstream clinical manifestations.

Cell types and processes involved: Primarily post-mitotic and developing CNS neurons (CL:0000540 neuron), oligodendrocytes for myelination (CL:0000128 oligodendrocyte), and retinal ganglion cells for optic atrophy (CL:0000740). Key GO biological processes: translational elongation (GO:0006414), guanine-nucleotide exchange / GEF activity (GO:0005085), regulation of transcription in response to stress / cellular response to heat (GO:0034605), and nervous system development (GO:0007399). Relevant GO cellular components: cytoplasm/polysome for the elongation function and nucleus (GO:0005634) for the eEF1BδL transcription-factor function. Affected anatomy (UBERON): corpus callosum (UBERON:0002336), cerebral white matter (UBERON:0002437), cerebellum (UBERON:0002037), optic nerve (UBERON:0000941), cerebral cortex (UBERON:0000956).

Genotype–domain relationship

Variant location	Example variants	Isoform/function affected	Reported families
Alternatively spliced long-isoform domain	prior truncating variants	eEF1BδL (heat-shock TF function)	2 prior families
C-terminal GEF domain	c.1215+1G>C, c.827_844del, c.1651A>T, c.966C>A	GEF (translation-elongation) function	2 new families (Averdunk 2023)

The convergence of both variant classes on the same severe phenotype argues that *EEF1D* is essential through **both** its translation-elongation and stress-response functions.

Evidence Base

PMID	Title	Evidence type	How it supports the findings
36576126	*Expanding the spectrum of EEF1D neurodevelopmental disorders: Biallelic variants in the guanine exchange domain*	Human clinical (case series)	Defining paper. Establishes *EEF1D*/eEF1Bδ as causal, documents homozygous truncating/GEF-domain variants, the GEF mechanism, isoform structure, and the full phenotype (F001, F002, F003, F006, F009)
30370994	*The role of translation elongation factor eEF1 subunits in neurodevelopmental disorders*	Review	Places *EEF1D* within the eEF1B GEF complex and the shared NDD pathway (EEF1B2, VARS), defining the molecular differential diagnosis (F002, F008)
25436608	*Characterisation of translation elongation factor eEF1B subunit expression in mammalian cells and tissues and co-localisation with eEF1A2*	In vitro / expression	Supports essentiality of eEF1B subunits; subunit ablation impacts viability/cycling, consistent with mouse null lethality (F007)
28097321	*Diagnostic Yield and Novel Candidate Genes by Exome Sequencing in 152 Consanguineous Families With Neurodevelopmental Disorders*	Human clinical (cohort)	Illustrates autozygosity-mapping + exome strategy used to discover recessive NDD genes (F006, F008)
21597468	*Transformation of eEF1Bδ into heat-shock response transcription factor by alternative splicing*	In vitro (molecular)	Establishes the eEF1BδL nuclear HSF1/Nrf2/HO1 stress-response function, explaining the second mechanistic arm (F005)

Supporting curated databases: **ClinVar** (variant list and classifications), **OMIM 621150** and **MONDO:0976263 / MedGen C6012708** (disease identity and nomenclature), **Monarch/HPO** (46 phenotype annotations), **gnomAD v2** (constraint metrics, F004), and **IMPC/MGI:1913906** (mouse null lethality, F007).

Section-by-Section Summary (knowledge-base fields)

- **Disease information:** Severe AR congenital NDD. OMIM 621150; MONDO:0976263; MedGen C6012708/1876520; acronym **NEDTCHAL**. Synonyms derive from the cardinal features (thin corpus callosum, hypotonia, absent language). Information is aggregated from disease-level resources plus the small published case series (individual patients). No distinct ICD-10/ICD-11 code; grouped under neurodevelopmental/intellectual-disability categories.
- **Etiology:** Monogenic; biallelic *EEF1D* LoF. No environmental or infectious cause. Principal non-genetic risk factor is **parental consanguinity** (increases homozygosity). No protective factors or gene–environment interactions established.
- **Phenotypes:** See Finding 3 and the HPO list; severe, congenital-onset, structurally static but with developmental stagnation; quality-of-life impact profound (non-verbal, non-ambulatory, tube-dependent).
- **Genetic/molecular:** Causal gene *EEF1D* (HGNC:3211; 8q24.3; OMIM gene 130592). Variant types: splice, frameshift, nonsense, in-frame truncating. Germline; homozygous. Functional consequence: loss of function (elongation GEF and/or long-isoform TF). No modifier genes or epigenetic mechanisms established; no large chromosomal abnormalities.
- **Environmental:** None implicated beyond consanguinity.
- **Mechanism:** See causal chain above. GO:0006414, GO:0005085, GO:0034605; CL:0000540, CL:0000128, CL:0000740.
- **Anatomy:** CNS primary — corpus callosum (UBERON:0002336), cerebral white matter (UBERON:0002437), cerebellum (UBERON:0002037), optic nerve (UBERON:0000941). Bilateral/symmetric. Subcellular: cytoplasm/polysome and nucleus (GO:0005634).
- **Temporal:** Congenital onset; structurally static malformations with developmental stagnation; chronic, lifelong; fetal/early-postnatal critical window.
- **Inheritance/population:** Autosomal recessive; ultra-rare; consanguinity-linked; presumed high/complete penetrance with variable expressivity; *EEF1D* not haploinsufficient (gnomAD). No founder mutation defined.
- **Diagnostics:** WES/WGS (first-line), autozygosity mapping, supportive brain MRI and EEG; no biomarker.
- **Prognosis:** Severe; poor functional outcome; morbidity/mortality driven by aspiration pneumonia and feeding failure.

- **Treatment:** Supportive only (gastrostomy, antiseizure medication, spasticity/respiratory care, rehabilitation). No NCIT-coded disease-specific therapy; supportive-care NCIT concepts include enteral feeding, anticonvulsant therapy, physical/occupational/speech therapy.
- **Prevention:** Genetic counseling, carrier/cascade testing, prenatal/preimplantation genetic diagnosis in known families.
- **Other species/models:** Mouse *Eef1d* (MGI:1913906; NCBI Taxon 10090) ortholog; constitutive null is embryonic/preweaning lethal (IMPC) — does not recapitulate the human NDD; conditional/hypomorphic models needed. No naturally occurring animal disease reported.

Limitations and Knowledge Gaps

1. **Very small evidence base.** Only ~4 families have been reported, so the phenotypic spectrum, penetrance, and natural history are incompletely defined, and the genotype–phenotype correlation (long-isoform vs GEF-domain variants) rests on few observations.
2. **No functional validation in patient neurons.** The causal chain from reduced elongation/stress response to specific brain malformations is mechanistically plausible and partly demonstrated in vitro, but not directly shown in patient-derived neurons or organoids.
3. **No faithful animal model.** The constitutive mouse null is lethal, so neurodevelopmental mechanisms cannot be studied in it; conditional or hypomorphic models do not yet exist in the literature reviewed.
4. **No biomarker or therapy.** Diagnosis is purely sequence-based, and no therapeutic target has been tested.
5. **Epidemiology unquantified.** No prevalence/incidence, sex ratio, or geographic-distribution data beyond the consanguinity association.
6. **Some HPO identifiers require confirmation** against the live ontology; the phenotypic concepts, however, are well supported.

Proposed Follow-up Experiments / Actions

1. **Patient-derived iPSC neurons/organoids** carrying GEF-domain and long-isoform *EEF1D* variants to measure global and transcript-specific translation (ribosome profiling),

proteostasis, and HSF1/Nrf2/HO1 stress-response capacity — directly testing the two-arm mechanistic model.

2. **Conditional / hypomorphic mouse models** (neuron- or progenitor-specific *Eef1d* knockdown) to bypass embryonic lethality and determine whether corpus-callosum thinning, microcephaly, and myelination delay are recapitulated.
3. **International genotype–phenotype registry / GeneMatcher recruitment** to expand the cohort, refine penetrance/expressivity, and test whether long-isoform vs GEF-domain variants differ clinically.
4. **Isoform-resolved expression mapping** in developing human brain (e.g., BrainSpan/ Human Cell Atlas) to clarify where and when eEF1BδL vs short isoforms are required.
5. **Biomarker exploration:** assess whether impaired HO1/Nrf2 stress response or elongation defects produce measurable cellular signatures usable diagnostically or as pharmacodynamic readouts.
6. **Proof-of-concept rescue** (e.g., ASO/splice-modulation or gene supplementation) in patient cells to evaluate feasibility of restoring eEF1Bδ function.
7. **Prospective natural-history documentation** (feeding, respiratory, seizure, developmental trajectories) to inform anticipatory supportive-care guidelines.

Prepared from 9 confirmed findings across 5 primary papers and curated database annotations (ClinVar, OMIM, MONDO/MedGen, Monarch/HPO, gnomAD v2, IMPC/MGI). Evidence types are indicated per claim: human clinical, in vitro, model organism (mouse), and computational/ database.
